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Java Coding_S3_100196

Module: Coding

Question 1:

Q6 - Basic Square Pattern

Write a Program to Print the following Basic Pattern?

If Input is 5 then Print

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

Constraints:

Input :- First Line of Input Consists of One Integer Value (Rows & Columns).

Output :- Print the Square Pattern.

Constraints :- Given Input Must be Greater than Zero or else Print **Invalid Input**.

Example:

Input 1 : 5

Output 1:

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

1 0 1 0 1

Input 2 : 6

Output 2:

1 0 1 0 1 0

0 1 0 1 0 1

1 0 1 0 1 0

0 1 0 1 0 1

1 0 1 0 1 0

0 1 0 1 0 1

Explanation:

Print a Square Pattern by taking One Integer Input

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	5	1 0 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 1 0 1
2	6	1 0 1 0 1 0 0 1 0 1 0 1 1 0 1 0 1 0 0 1 0 1 0 1 1 0 1 0 1 0 0 1 0 1 0 1

Question 2:

Q3 - Sum of All Odd Numbers From an Array

Write a Program to Find the sum of all odd numbers from a Given Array?

Constraints:

Input :- First Line of Input Consists of One Integer Value (Array Size).

Second Line of Input Consists of few Integer Values Separated By Spaces (Array Elements).

Output :- Print the Sum of All the Odd Array Elements.

Constraints :- Array Size must be Greater then or equal to 5 or else Print "**InValid Array Size.**".

If there is no Odd Elements in a Given Array then Print "**No Odd Numbers.**".

Example:

Input 1 : 8

8 6 3 53 93 84 210 294

Output 1: 149

Input 2 : 3

1 2 3

Output 2: InValid Array Size.

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	8 8 6 3 53 93 84 210 294	149
2	3 1 2 3	InValid Array Size.

Question 3:

Artifact Alignment Challenge

In a futuristic lab, scientists are trying to analyze a set of unique artifacts. Each artifact is represented by a distinct weight, and the artifacts must be aligned in increasing order of their weights for a

precise experiment. Due to the fragility of the artifacts, they can only be flipped in batches from the
beginning up to a selected position in the line, meaning the order of the first k artifacts can be reversed in
one operation.
Your task is to reorder the artifacts by performing the minimum number of valid flips. A flip is valid if you
select an integer k (1 ≤ k ≤ number of artifacts) and reverse the order of the first k artifacts. The goal is to
ensure the artifacts are arranged in ascending order, while keeping the number of flips as low as possible.

Constraints:

- 1 ≤ number of artifacts ≤ 100
- 1 ≤ weight of each artifact ≤ number of artifacts
- All weights of artifacts are unique (i.e., the artifacts are a permutation of integers from 1 to number of
artifacts)

Example:

Example 1:

Input: 5 3 1 4 2
Output: [5, 2, 4, 3, 2]

Example 2:

Input: 4 1 2 3
Output: [4, 3]

Explanation:

The challenge is to repeatedly flip portions of the array to eventually get it sorted in ascending
order. The goal is to find the sequence of flips (the values of k you choose) that will sort the array in the
fewest steps possible. After sorting, you must return the sequence of flip operations used. TThe key is to
devise a strategy that sorts the array efficiently using no more than 10 * arr.length flips, and the output
should detail the exact flip sequence that leads to the sorted array.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	3 5 1 3 5 1	[2, 6, 2, 5, 4, 2, 3, 2]
2	2 1	[2]

Question 4:

The Quest for the Triad of Harmony

In the enchanted land of Numina, a mystical forest is home to various magical creatures, each
possessing a unique power represented by an integer value. However, the balance of the forest is
threatened as a dark force seeks to disrupt the harmony among the creatures. To restore balance, the wise
sorcerer Aeloria has devised a challenge for the brave adventurers of Numina.
Your task is to find all the triplets of creatures whose combined powers equal zero. Specifically, given an
array of integers representing the powers of the creatures, return all unique triplets
[powers[i],powers[j],powers[k]]such that i≠j, i≠k,j≠k, and
powers[i]+powers[j]+powers[k]=0.

Constraints:

The enchanted powers array must meet the following criteria:
- The length of powers must be between 3 and 3000.
- Each creature's power must be within the range of -10^5 to 10^5.

Example:

Example 1:

Input: 5
-1 0 1 2 -1 -4
Output: [[-1, -1, 2], [-1, 0, 1]]

Example 2:

Input: 6
-3 -1 0 2 4 5
Output: [[-3, -1, 4]]

Explanation:

The task is to identify all unique triplets of magical creatures in the enchanted land of Numina
whose combined powers equal zero.By executing the findTriplets function, adventurers will be able to
uncover these unique alliances among the magical beings, thereby contributing to the restoration of balance
in the mystical forest of Numina. This task not only tests analytical skills but also emphasizes the
importance of collaboration among creatures in their fight against the dark forces threatening their harmony.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	4 1 2 -3 -2	[[-3, 1, 2]]
2	4 1 2 -2 -1	[]

	1 2 -2 -1	